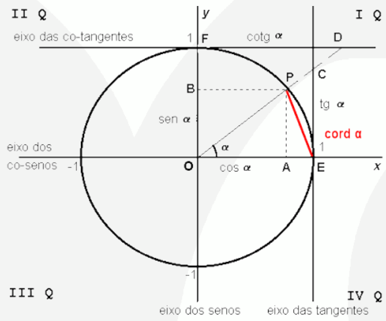
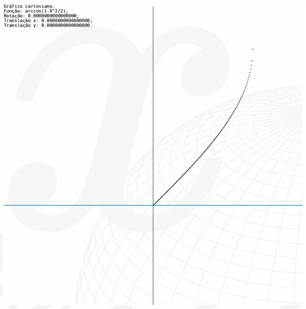
A inversa, a derivada, e a integral da função corda.

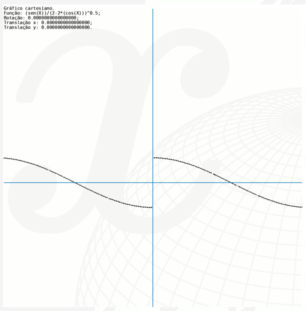


$$cord\ \alpha = \sqrt{2(1-\cos\alpha)}$$

Inversa: seja $\begin{array}{c} arccord : [0,2] \to [0,\pi], \\ x \mapsto arccord\ x \end{array}$ $\boxed{arccord\ x = \arccos\left(1-\dfrac{x^2}{2}\right)}$.



Derivada: $\boxed{(cord\ \alpha)' = \dfrac{\sin\alpha}{\sqrt{2-2\cos\alpha}}}$.



Observemos que, para $0 \leq \alpha \leq 2\pi$, $cord\ \alpha = 2\sin\dfrac{\alpha}{2}$.

Logo,

$$\boxed{\int cord\ \alpha\ d\alpha \ = \ -4\cos\frac{\beta}{2} + c,\ \ \alpha = 2k\pi + \beta,\ \ k \in \mathbb{Z}, 0 \leq \beta < 2\pi}.$$

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Sugestões, comunicar erros: ”a.vandre.g@gmail.com”.